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AI skills supply and demand

An analysis through online job advertisements and education and training offer

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Contents

Abstract

This report provides evidence to inform the European Commission's *Apply AI Strategy* — announced in the AI Continent Action Plan and launched to enhance AI adoption in strategic sectors — by analysing the alignment between AI-related education and labour market demand in the EU. Using Studyportals data (2024–2025) and WIH-OJA online job advertisements (2020–2023), it examines master's and short courses alongside job advertisements for ICT specialist occupations. The education and training offer's content is dominated by Machine Learning, especially in the case of short courses, with notable attention to AI ethics, while generative AI remains marginal. Most programmes that feature AI topics are offered in the ICT field, but relevant provision also exists in Engineering, Business, administration and law, and other disciplines, albeit with low penetration. AI-related job demand is highly concentrated in software and applications developers and analysts occupations (62% of AI-related OJAs) and database and network professionals, with strong AI specialisation in systems analysts. The most in-demand job profile in AI-related OJAs is the AI/ML Engineering, appearing in almost one-third of the job advertisements that explicitly mention AI job profiles, followed by Data Analysis, Data Engineering, and AI/ML Development. Together, these four profiles account for 98% of all job descriptions referring to AI roles. While AI-related education and training offer and ICT specialist jobs demand are broadly aligned, limited AI integration in the academic offer of non-ICT fields and in emerging technologies may lead to future skills gaps in some sectors.

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Executive summary

Policy context

The reinforcement of the workforce's competencies in artificial intelligence (AI) is among the objectives of several European Union's policies and regulation, such as the AI Continent Action Plan, the AI Act, or the Digital Decade Policy programme. This report provides information and evidence in relation to AI-relevant education and training offer, and AI-related labour market demand. Specifically, it examines master's and short courses offered in the 2024-2025 academic year across various fields of study (including illustrative examples) alongside online job advertisements for ICT specialists from 2020 to 2023. The overall aim of this exploratory study is to describe the existing AI-related education and training offer and skills demand, assessing the overall trends and overall alignment, and observe the evolution of most-demanded AI profiles. This would allow for the identification of potential priorities to help ensure a competitive, inclusive, and trustworthy AI ecosystem in the EU. Findings are highly relevant to skills and innovation policies, and sectoral strategies where AI is critical. Results are timely to inform upcoming policy actions on AI adoption, workforce preparedness, and micro-credential frameworks.

Key conclusions

Education provision broadly aligns with labour market demand in AI subdomains. The results show some potential education fields where AI programmes could be reinforced. For example, AI-related education and training offer is highly concentrated in ICT fields, with low presence in non-ICT fields (e.g. health, agriculture, social sciences), risking AI skills shortages in some critical sectors. Both master's programmes and short courses focus on Machine Learning, and AI in general. Short courses tend to be more specialised, offering potential for rapid reskilling or upskilling to meet specific needs. AI ethics and, to a minor extent, Generative AI, are emerging in curricula during 2024-2025 but scarcely appeared in job advertisements during 2020-2023. Notwithstanding the time misalignment between both sources, this suggests that education and training offer is anticipating these aspects of AI skills—and that proactive policy integration might anticipate future demand. AI-related job demand in ICT specialist occupations is concentrated in software, data, and systems analysts roles, with much lower concentrations in technician roles. Results suggest that policy measures should accelerate AI integration across disciplines and expand short-course and micro-credential availability. Also, AI-related jobs demand, whose analysis could be extended to a wider range of occupations, should monitor trends in generative AI and ethics alongside other AI-related skills to check and address potential imbalances in this regard. The evidence confirms the value of examining AI-related education and training offer with labour market demand using a harmonised approach to both data sources, and provides a way to identify potential sectoral and thematic imbalances for policymakers to consider.

Main findings

— **AI-related course offerings**: Master's programmes in AI are concentrated in the subdomains of Machine Learning (32%), Generic AI (26%), Robotics & Automation (13%), and AI Ethics (10%), while coverage of Generative AI remains very limited (1%). **Short courses** follow similar patterns, with Machine Learning (42%) and Generic AI (27%) dominating, and smaller shares for Robotics & Automation (4–5%), AI Ethics (4–5%), and Generative AI (2%). These courses are generally more specialised than master's programmes and are well suited to rapid reskilling and upskilling to meet emerging digital technological developments and sectoral needs.

— **Distribution of AI-related course offerings across fields of study**: Almost half (47%) of AI-related master's degrees are offered within the field of ICT, followed by Engineering, manufacturing and construction (19%), Business, administration and law (13%), and Natural sciences, mathematics and statistics (11%). The AI penetration rate, or share of AI-related master's degrees to all master's offer, is highest in the field of ICT (44%), with much lower integration in other fields.

— **The distribution of AI programmes across disciplines** shows that while ICT is most often offering specialised programmes in AI, other fields such as Business, administration and law and Engineering, manufacturing and construction tend to offer broader programmes, underlining the need for multidisciplinary profiles that combine AI expertise with sector-specific knowledge.

— **AI-related demand in online job advertisements of ICT specialists**: AI-related online job advertisements (OJAs) are dominated by software and applications developers and analysts (62%) and database and network professionals (16%). Three occupations show a higher level of AI specialisation —i.e., showing higher concentration among AI OJAs than among overall ICT specialists OJAs— Systems Analysts; Database and Network Professionals (not elsewhere classified, n.e.c.); and Software and Applications Developers and Analysts. In contrast, AI specialisation is lowest among technician and sales professional occupations. Results suggest a marked unevenness in the extent to which different ICT specialist occupational advertisements contain AI-related themes. Results also show that the most demanded AI profiles (roles) across all ICT specialist occupations are AI/ML engineering, AI/ML development, Data analysis and Data engineering.

— **Comparisons of AI-related education and training offer and job demand**: A broad comparison of the distribution of AI-related education and training offer and AI-related job demand for ICT specialists by AI subdomains suggests that offer and demand align quite closely, with some exceptions. For example, AI ethics appears in the education and training programmes studied but is absent from OJAs. This could be in part because the data on programmes cover a more recent period than that on job demand, or because the analysis is limited to ICT specialists occupations, which may not be the most common ones for ethicists.

Related and future Joint Research Centre work

This study complements JRC's AI Watch and prior analysis of education and training offer on advanced digital domains, and is a part of its digital skills monitoring work. Future work that builds on the present analysis could extend the examination of AI-related demand beyond ICT to a wider range of occupations, particularly those of strategic or critical importance, such as mobility and automotive, energy, environment and climate, and the cultural and creative industries (including media).

Quick guide

This study maps the EU's AI education (master's, short courses) against labour market demand (online job ads), using keyword-based thematic analysis. It identifies alignment, gaps, and policy levers which can be considered to strengthen AI skills supply for the EU's strategic, ethical, and competitive AI adoption.

1. Introduction

Artificial Intelligence (AI)—including generative AI—is profoundly shaping society and the economy, giving rise to multiple, complex opportunities and challenges (Abendroth Dias et al., 2025), including in the labour force. For example, a recent survey on AI skills in the workplace (Cedefop, 2025) indicates that use of AI systems in work across 11 EU countries is both common (with 28% of workers using these systems), and uneven (ranging from about 15%-42%). Also, while 44% of workers in the study reported a need to develop knowledge and skills in relation to AI systems, only 15% had participated in training related to AI in the past year.

Various European Union initiatives and regulations—such as the Digital Decade Policy Programme[1] , the AI Act[2] and the AI Continent Action Plan[3]—are responding to these changes. Among other ambitious objectives, they seek to enhance AI competences within the workforce. Specifically, the *AI Continent Action Plan's* AI talent base domain aims to reinforce AI skills, including basic AI literacy and diverse talent, throughout the EU. Actions include an increase in the overall provision of EU bachelor's and master's degrees and PhD programmes in key technologies, including AI. A key action in this domain is the establishment of a one-stop AI Skills Academy, which will offer education and training programmes on AI, in particular generative AI. It will upskill and reskill students and professionals in key sectors and develop a pilot generative AI-focused degree. This will enable top-level experts in generative AI to educate and train the AI Skills Academy's students, while advancing their own research in the field. The Academy will also pilot an AI apprenticeship programme and develop 'scholarship' and 'returnship' schemes to attract more women to the field of AI.

The *Apply AI Strategy*, announced in the AI Continent Action Plan and published in October 2025, serves as a blueprint for the further adoption and integration of AI in EU's strategic sectors, such as healthcare, mobility and the public sector. In particular, it aims to enable European companies to be global AI front-runners, to increase the quality of services in the public sector and foster the integration of AI technologies into strategic sectors.

Within this broader context, this report provides scientific evidence on the AI-related education and training offer and demand for AI-related jobs. This analysis describes AI (including generative AI)-related education and training offer (concretely, masters' programmes and short professional courses) and demand (in online job advertisements, OJAs).

A comparison of offer and demand provides preliminary strategic insights into potential gaps or supply-demand mismatches. The analysis not only quantifies the offer and demand aspect, but also explores thematic dimensions, presenting the main characteristics of offer and demand in what regards their AI components.

The methodology employed offers a way to monitor these issues while the implementation of the AI Continent Action Plan (and Apply AI Strategy) progresses, as well as insights for potential actions by policymakers.

1 https://digital-strategy.ec.europa.eu/en/library/digital-decade-policy-programme-2030

2 https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

3 https://digital-strategy.ec.europa.eu/en/library/ai-continent-action-plan

2. Methodology

2.1. AI-related education and training offer

Analysis of education and training offer is based on data from Studyportals[4], which has extensive global and European coverage, particularly for English-language programmes. The dataset's richness in programme details enables nuanced analysis of programme characteristics. A notable caveat is the exclusion of non-English programmes, which may limit representativeness in regions where local languages dominate. However, previous research (López-Cobo et al., 2019) indicates that the dataset at master's level for technological domains is representative.

The work follows the methodology developed in López-Cobo et al. (2019) and revised in Righi et al. (2020). A keyword-based text-mining approach is used. Semi-automated keyword lists, derived from scientific publications and prior studies, were used to flag relevant programmes (see the full list of keywords in Table A1 in Annex 1).

For this analysis, education programmes are categorised as "specialised" or "broad" based on the intensity of their focus on AI. Specialised programmes emphasise AI as their central theme (e.g., a Master's degree in Natural Language Processing), whereas broad programmes incorporate the domain as a secondary or supplementary component within a more general curriculum (e.g., a Master's degree in biomedicine that includes a foundational course on AI, or a degree in Engineering that includes a course on computer vision). Operationally, "specialised" programmes are those where AI-related keywords are central—appearing in the title or short description, or including at least three different keywords in the long description of the programme.

The results provide an overview of the AI-related education and training offer in AI in master's and short courses across EU universities in the academic year 2024-2025. The focus on master's programmes is relevant, as they frequently mark the final stage of education and are more commonly offered in English than lower educational levels[5]—aligning with the scope of the available data, which consider only English-language programmes. Short professional courses, designed for professionals and graduates seeking to expand their expertise in specialised fields and diversify their skill sets, are gaining traction, which is partly driven by the widespread accessibility of massive open online courses (MOOCs) that offer flexible learning options[6].

[4] Studyportals is a platform covering over 150,000 academic programs from 3,700 universities across 120 countries, including 50,000 European programs, updated yearly.

[5] https://monitor.icef.com/2024/09/european-study-destinations-now-offering-thousands-of-english-taught-degree-programmes/

[6] These courses are typically delivered via online platforms like Coursera, edX, and Udemy, which aggregate offerings from universities and educational networks. Additionally, many are directly provided by prestigious institutions such as Utrecht Summer School and TU Delft in the Netherlands, TU Berlin in Germany, and Audencia Business School in France. The duration of these programs generally spans from a few days to several months, catering to diverse learning goals and schedules.

2.2. AI-related labour market demand

The analysis of AI labour market demand uses the European Commission's WIH-OJA NLP (Web Intelligence Hub Online Job Ads, supported by Natural Language Processing) database[7]. The data contains detailed information on the jobs and skills that employers seek based on online job advertisements (OJAs) (Napierala, 2024). It includes 32 European countries and millions of OJAs collected from thousands of sources, including private job portals, public employment service portals, recruitment agencies, online newspapers and corporate websites.

The present analysis was carried out with the support of Cedefop[8] using a keyword-based query search approach on OJAs descriptions. The same keywords used in the analysis of education and training offer (Annex 1, Table A 1) were used, and supplemented with an additional set of AI job profile keywords (Annex 1, Table A 2) meant to capture AI roles typically used in job descriptions. This additional set of keywords was developed through a review of several online descriptions of AI profession roles and was included to ensure a more complete coverage of the overall domain[9]. AI profiles are distinct from ICT specialist occupations in that they cover specific roles within occupations, and they can appear across multiple occupations. In this report, we define AI-related OJAs as those whose description contain one or more keywords (Annex 1).

The reference period of the OJAs data is 2020-2023[10], and the occupations included are ICT specialists, i.e., workers who can develop, operate and maintain ICT systems, and for whom ICT constitutes the main part of their job[11]. The data in the present analysis consists of a representative sample of OJAs in the 27 EU Member States.

When examining occupations, we have reported the results at the most detailed (four-digit) level of the International Standard Classification of Occupations (ISCO)[12], with a small number of exceptions. To assure robustness in the results, we have combined occupations for which we identify less than 50 AI-related OJAs across the years 2020-2023 combined. To do so, we aggregate them into their 3-digit parent ISCO group (when all 4-digit occupations of the group are considered as ICT specialists). Alternatively, if the number of AI OJAs for the period falls below 50 and the occupation cannot be grouped under the 3-digit parent group, it is left out of the analysis. Therefore, the analysis is performed at 4-digit level of the ISCO classification except in the following cases:

- Occupations *3521 - Broadcasting and Audio-visual Technicians* and *3522 - Telecommunications Engineering Technicians* are aggregated into *352 - Telecommunications and Broadcasting Technicians*.
- Occupations *7421 -- Electronics Mechanics and Servicers* and *7422 - ICT Installers and Servicers* and are aggregated into *742 - Electronics and Telecommunications Installers and Repairers*.

[7] https://cros.ec.europa.eu/wih/oja

[8] Cedefop extracted and provided the relevant data to the JRC on the basis of a list of keywords and specifications supplied by the JRC.

[9] The review of online sources progressed until reaching 'saturation' (i.e. when no new profiles were being found).

[10] At the time of the data extraction (April-May 2025), the data for 2024 onwards was not yet complete or available.

[11] Eurostat and the OECD define ICT specialists and propose a statistical definition of ICT specialists based on the International Standards Classification of Occupations (ISCO) 2008 (OECD, 2015).

[12] https://ilostat.ilo.org/methods/concepts-and-definitions/classification-occupation/

- Occupation *2356 - Information Technology Trainers* cannot be grouped under occupation *235 - Other Teaching Professionals* as the rest of 4-digit occupations in 235 are not ICT specialists. Therefore, this occupation is not considered in the analysis.

Some caveats should be borne in mind in interpreting OJAs results. Positive and ‘soft’ skills tend to be overly emphasised due to social desirability bias (Fernandez-Macias & Sostero, 2024), and the present analysis is restricted to the English language[13]. Also, the extracted data available to the JRC does not contain the job description’s text, therefore does not allow a view of a keyword instance within the broader context of a ‘chunk’ of source text. This means that it has not been possible to undertake a contextual verification of keyword matches. To reduce as much as possible the risk of false positives, i.e., the wrong identification of an OJA as AI-relevant, for most keywords the search algorithm forces the co-occurrence of such keyword with “artificial intelligence” or “AI”.

2.3. Comparative analysis

The use of a common list of keywords to identify relevant education programmes and OJAs allows us to perform a preliminary comparative analysis based on their thematic content. The keywords are clustered using a taxonomy of AI subdomains, developed by Samoili et al. (2020) in the framework of AI Watch, and complemented with an additional AI subdomain “generative AI” to account for recent developments in the area.

Comparing the distributions of AI subdomains across education and training offer and labour market demand provides insights about the alignment or misalignment between what is being sought in the labour market and the education and training offer.

It should be noted that the timeframe of the two data sources is not the same: Education and training offer data covers 2024-2025, while OJAs cover 2020-2023.

[13] While many keywords in digital technological contexts are language neutral (e.g., names of software), not all are. Therefore, it is possible that the analysis underestimates the total share of AI-related OJAs.

3. Results and analysis

3.1. AI-related offer in Master's programmes and short professional courses

This section describes the main characteristics of the education and training offer considered in the study. and **Figure 2** provide an overview of the thematic content of the education and training offer in AI across EU universities in the academic year 2024-2025 for master's programmes and short courses, respectively. The thematic content of a programme is identified by clustering AI-related keywords that fall within the same subdomain, based on an established AI taxonomy (see Annex 1). **Figure 1** shows that **master's-level AI programmes in the EU are primarily concentrated on Machine Learning (32%) and general AI topics (26%),** reflecting their foundational role in academic training. **Generative AI, though still an emerging field, represents only 1%** of AI master programmes. AI Ethics constitutes 10% of the content of AI master's, pointing to an alignment of education offer with evolving knowledge and skills requirements to support critical and ethical design and deployment of AI systems.

Figure 1 differentiates between *specialised* and *broad* programmes[14]. Overall, the data does not reveal major differences in thematic coverage between specialised and broad programmes, suggesting a relatively consistent distribution of AI topics across both types of educational offers. However, some small deviations are visible across content areas. For instance, AI Ethics and AI Applications show a relatively higher share of specialised programmes, likely reflecting a deliberate focus on domain-specific implementation of AI solutions or on addressing the ethical challenges posed by AI technologies. Conversely, in Robotics & Automation, broad programmes are more common than the specialised ones, which may suggest that AI is often treated as a supporting technology within broader robotics curricula rather than as the primary subject of study.

Figure 2 presents a similar categorisation for short courses, which generally run from a few days to a few months, and are typically designed to complement formal education and to support upskilling and reskilling of the workforce through concise and targeted content. These courses, due to their flexible format, may also lead to micro-credentials[15]; however, whether the short courses in the dataset result in micro-credentials is not explicitly recorded, so this cannot be determined with certainty. In line with the European Commission's *Digital Education Action Plan* (2021–2027)[16], micro-credentials play a key role in enabling flexible, targeted learning opportunities through short courses, helping learners quickly acquire specific skills that match labour-market needs by certifying the learning outcomes (Sanchez-Barrioluengo et al., 2025). For this reason, they are often promoted

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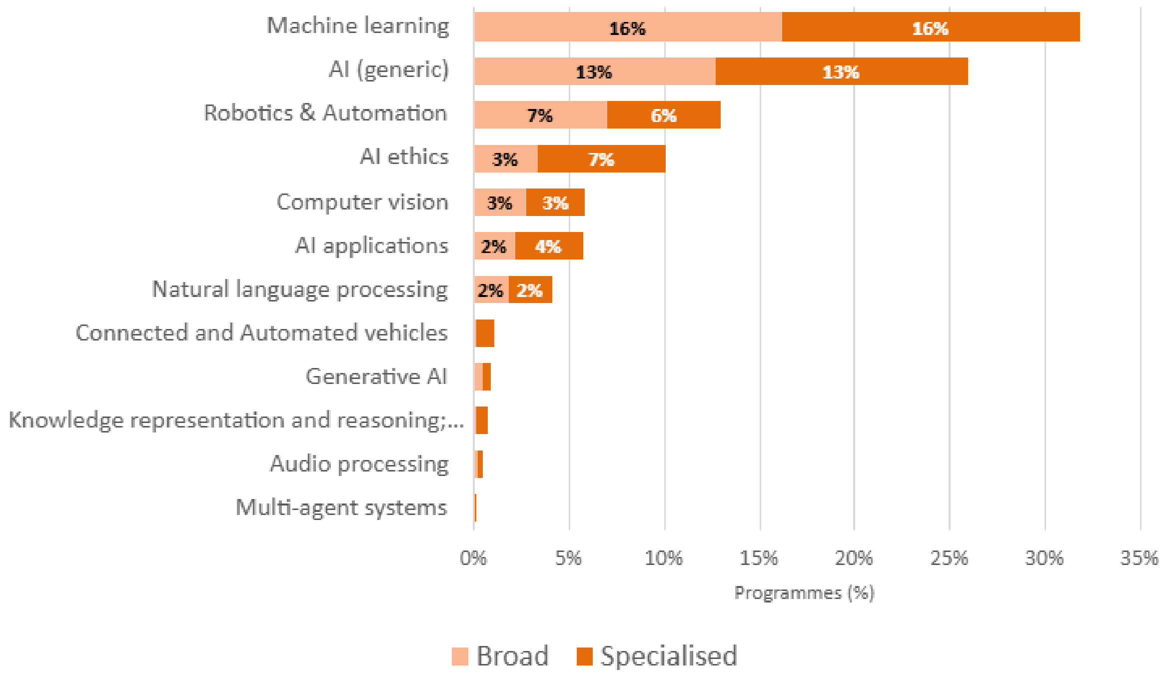
14 As explained in section 2, specialised programmes centre around AI as their primary focus (e.g., a Master's in Natural Language Processing), whereas broad programmes integrate AI as a secondary or supplementary element within a more general academic framework.

15 Micro-credentials are defined as: "the record of the learning outcomes that a learner has acquired following a small volume of learning. These learning outcomes will have been assessed against transparent and clearly defined criteria. [...] Micro-credentials are owned by the learner, can be shared and are portable. They may be stand-alone or combined into larger credentials. They are underpinned by quality assurance following agreed standards in the relevant sector or area of activity" (Council Recommendation - Higher Education Consultation Group, 2022, p. 13).

16 https://education.ec.europa.eu/focus-topics/digital-education/actions

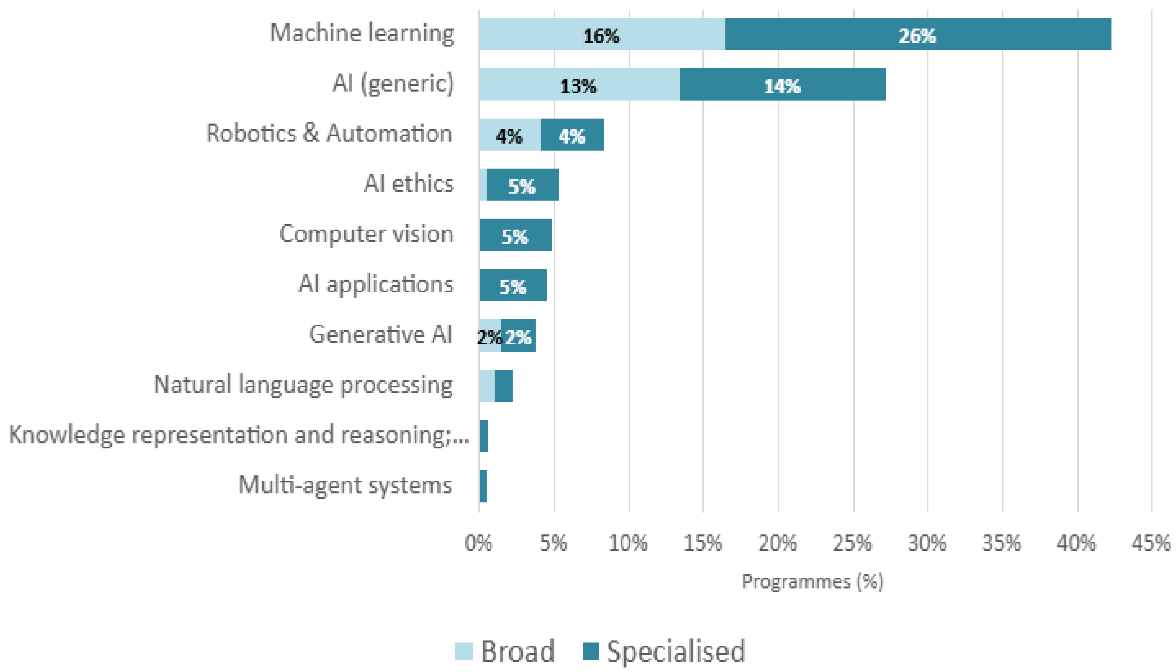
as an efficient way to upskill workers across the lifespan, particularly when traditional degree programmes may be less accessible or practical (Sanchez-Barrioluengo et al., 2025).

Figure 1. Distribution of AI master's degrees by content area, EU 2024-2025



Source: JRC's elaboration based on Studyportals data.

Figure 2. Distribution of AI Short courses by content area, EU 2024-2025



Source: JRC's elaboration based on Studyportals data.

Importantly, short courses can be easier to update and adapt to emerging technologies and evolving labour market needs, given their modular structure, shorter duration, and less burdensome administrative processes. This could make them particularly suited to respond to rapid technological change and would appear to offer a stronger lifelong learning and workforce adaptability in the AI domain.

The data indicate that **short courses are primarily focused on Machine Learning (42%) and AI (generic) topics (27%),** together accounting for nearly 70% of the total offer. This strong emphasis highlights the foundational and transversal nature of these areas, which likely appeal to a wide audience of learners seeking either an introduction to AI or upskilling in essential techniques. Unlike master's programmes, short courses appear more sharply concentrated around a few core themes, with limited diversification into niche or emerging fields. For instance, Robotics & Automation, AI Ethics, Computer Vision, and AI Applications each account for only 4–5% of programmes. On the other hand, **Generative AI remains marginal, at just 2%** of the short course provision. Despite its rapid development and visibility, this figure may indicate a lag between technological innovation and the integration of such topics into continuing education formats.

Similarly, **Figure 2** highlights the differentiation between specialised and broad programmes. In the case of short courses, specialised programmes are more widespread than in master's programmes, reflecting the inherently targeted nature of short courses, which are often designed to focus on specific AI topics. This is particularly evident in areas such as AI Ethics, Computer Vision, and AI Applications—where nearly all offerings are specialised. This trend mirrors what is observed in master's programmes but is even more pronounced. Additionally, short courses show a strong concentration of specialised offerings in Machine Learning, highlighting a clear demand for focused, skills-based training in rapidly evolving AI subfields.

Figure 3 and **Figure 4** illustrate **the distribution of AI-related programmes across fields of education** or discipline in which the programme is taught[17] (blue bars), and the penetration rate of the AI-related offer within each field, which refers to the proportion of AI programmes relative to the total number of programmes in that specific field (red circles). This analysis shows how AI programmes are integrated across different disciplines, providing AI-related knowledge and skills for future workers who will apply their knowledge in a wide range of occupations and sectors of the economy.

Figure 3 shows, as expected, that the **field Information and Communication Technologies (ICT) offers the highest proportion of AI master's degrees** (47% of all AI offer (blue bars)), followed by Engineering, manufacturing and construction (19%); Business, administration and law (13%); and Natural sciences, mathematics and statistics (11%). It is interesting to note that Arts and humanities, a less traditional technical field, offers 4% of all AI-related master's degrees.

As reported by **Figure 3**, European universities offer a wide range of AI-focused programmes across disciplines. For instance, In ICT, Spain's *Master in Big Data and AI Solutions* and Italy's *Human-Centred AI* equip students with skills in generative AI, data science, and ethical human-AI interaction. In engineering, Germany's *Applied AI in Automotive* provides flexible, online training for

[17] According to the Fields of education and training 2013 (ISCED-F 2013) classification. The International Standard Classification of Education (ISCED) is a framework for assembling, compiling and analysing cross-nationally comparable statistics on education. The 2013 revision focused on the fields of education and training (ISCED-F).

AI applications in the automotive sector. France's *AI for Marketing Strategy* MSc combines business and technical skills to enhance marketing through AI tools. In health and welfare, programmes like *AI for Health* (Netherlands) and *Medical Image Computing* (Spain) prepare students for roles in AI-assisted diagnostics and medical imaging. The social sciences and arts and humanities are covered by degrees such as *AI – Cognitive Science* (Netherlands) and *Visual Computing and Game Development* (Czech Republic), focusing on cognitive modelling and intelligent systems. In services, France's *Smart Mobility* programme trains students to apply AI in transportation, while in agriculture, *Agricultural and Food Data Management* (France) applies AI and machine learning to precision farming.

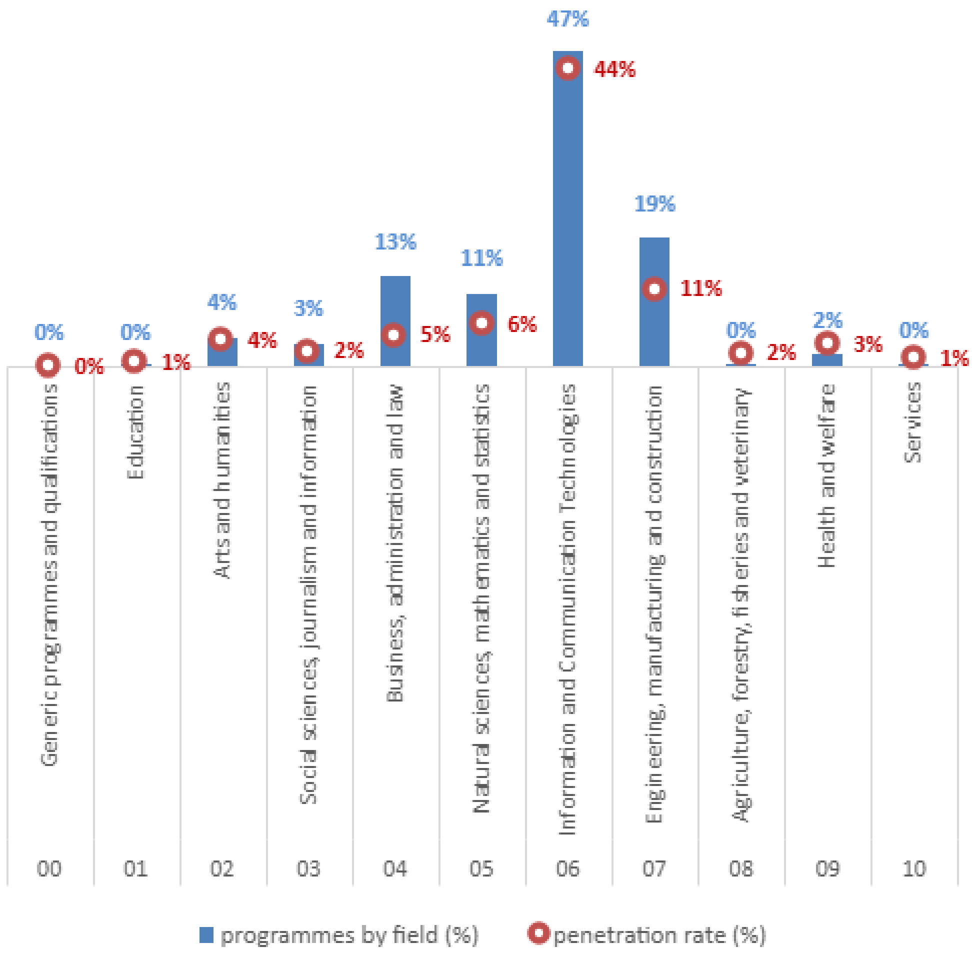
When looking at the penetration rate (red circles in **Figure 3**), the **ICT field also shows the highest penetration rate at 44%,** i.e., this is the proportion of all ICT master's degrees including AI content. This highlights AI's strong foundation in computer science and digital technologies. **This is followed by Engineering, manufacturing and construction at 11%, reflecting the growing role of AI in automation, robotics, and industrial innovation**. Moderate penetration rates (around 4–6%) are observed in other STEM fields such as Natural sciences, mathematics and statistics; as well as in Arts and humanities; and Business, administration and law, indicating an expanding but uneven integration of AI beyond its technical origins. In contrast, AI remains marginal in more applied or less technology-centric areas like Health and welfare, Social sciences, Agriculture, Education, and Services (1–3%). The much lower penetration rates of AI-related academic offer in these latter fields may indicate a time-lag between the growing societal and economic relevance of AI in these fields and their current low integration into courses.

As reported in **Figure 4 short courses with AI content are much more concentrated in the ICT field (59%** of all AI short courses are in this field), **followed by Business, administration and law (12%) and Engineering, manufacturing and construction (10%).** A relevant fourth position is occupied by the Health and welfare field (8%), which also shows the second highest penetration rate, with 6% of all short courses on health and welfare including some AI-related content. This highlights the growing interest of this application field for AI. The highest penetration rate is associated with ICT as for master courses. It is interesting to note that in short courses related Engineering, manufacturing and construction the penetration rate of AI is much lower than for the Masters courses. This suggests that reskilling and upskilling of jobs related to this field might require further attention to ensure that AI is covered in such short courses.

Some examples of short courses illustrate the wide variety of intensive, AI-focused learning opportunities available across Europe, including summer schools, professional training, and specialised modules. These are often designed for undergraduate students, early-career professionals, or PhD researchers, with a strong emphasis on practical, sector-specific applications. In the ICT field, Germany's *Artificial Intelligence in Industrial Applications* programme provides a comprehensive introduction to industrial AI, covering cyber-physical systems, sensor networks, and data integration. It also includes hands-on training in widely used tools aimed at those interested in AI applications across industrial environments. In France, CY Tech's *Summer School on Visual Computing & Artificial Intelligence* introduces undergraduate and graduate students to 3D rendering, computer vision, and deep learning, combining technical and AI content with cultural and language immersion. In Engineering, Germany's *Co-benefits from AI in Renewable Energy* and *Green Hydrogen and Power-to-X Professional* programmes target professionals, offering applied knowledge on AI's role in energy systems, project bankability, and hydrogen technologies. In Business, administration and law, the *ETGN Summer School on Agile Governance* (Italy) explores AI's influence on leadership, innovation, and accountability, while the *Digital Constitutionalism at the*

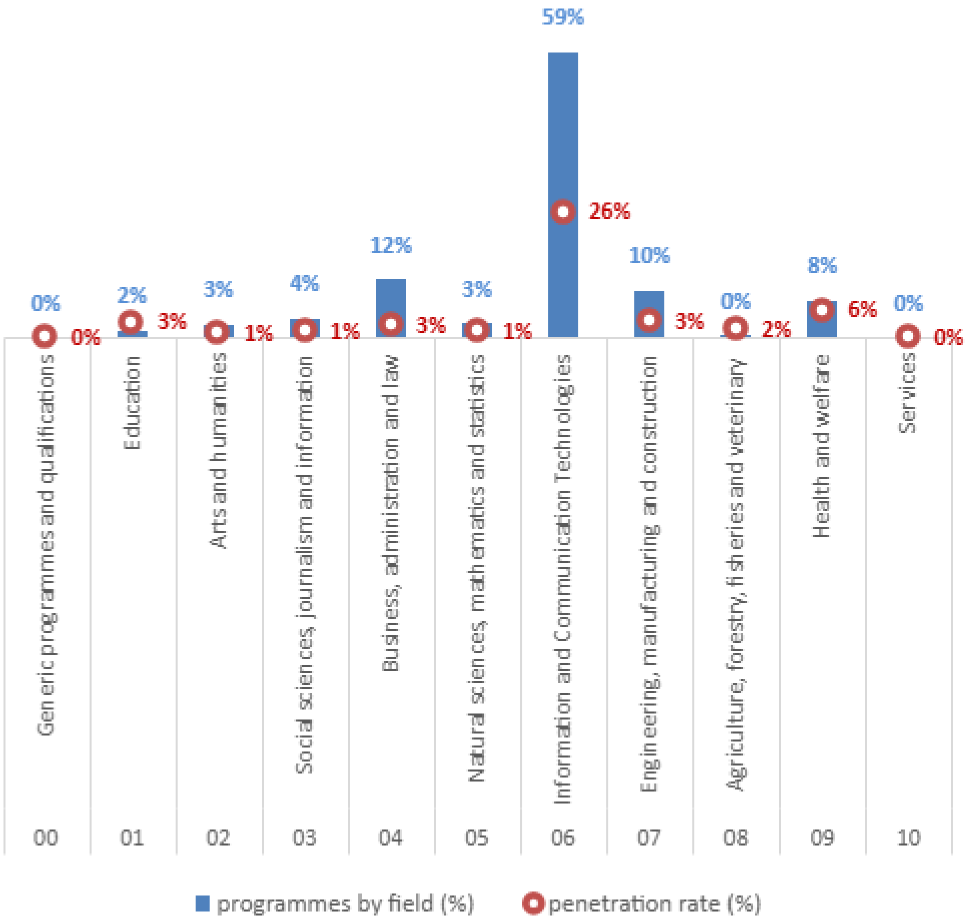
Age of AI course (Italy) brings together academics and practitioners to examine how AI challenges fundamental rights and democratic principles. In the Health domain, Sweden's *Introduction to Medical Image Analysis* focuses on computational and AI techniques for solving real-world medical imaging problems. Finally, in the Arts and humanities, the *Digital Tools for Humanists* summer school (Italy) offers a hands-on introduction to AI and digital tools for scholars in disciplines such as history and literature, delivered both on campus and online.

Figure 3. Distribution (blue bars) and penetration rate (red circles) of AI master's programmes by field of study (%), EU 2024-2025



Source: JRC's elaboration based on Studyportals data.

Figure 4. Distribution and penetration rate of AI Short courses by field of study (%), EU 2024-2025



Source: JRC's elaboration based on Studyportals data.

Figures A1 and A2 (see Annex 2) show the distribution of master's and short courses programmes by field of study and scope, distinguishing between specialised and broad AI programmes. Unsurprisingly, both graphs in Annex 2 indicate that programmes within the ICT field are more often specialised in AI. In contrast, fields such as Business, administration and law; and Engineering, manufacturing and construction; show a higher share of broad AI programmes, likely reflecting the role of AI as a supporting tool rather than the central focus in these disciplines. This highlights the importance of developing programmes that foster multidisciplinary profiles, equipping AI specialists with expertise in specific sectors in order to facilitate the development of sector-relevant AI applications.

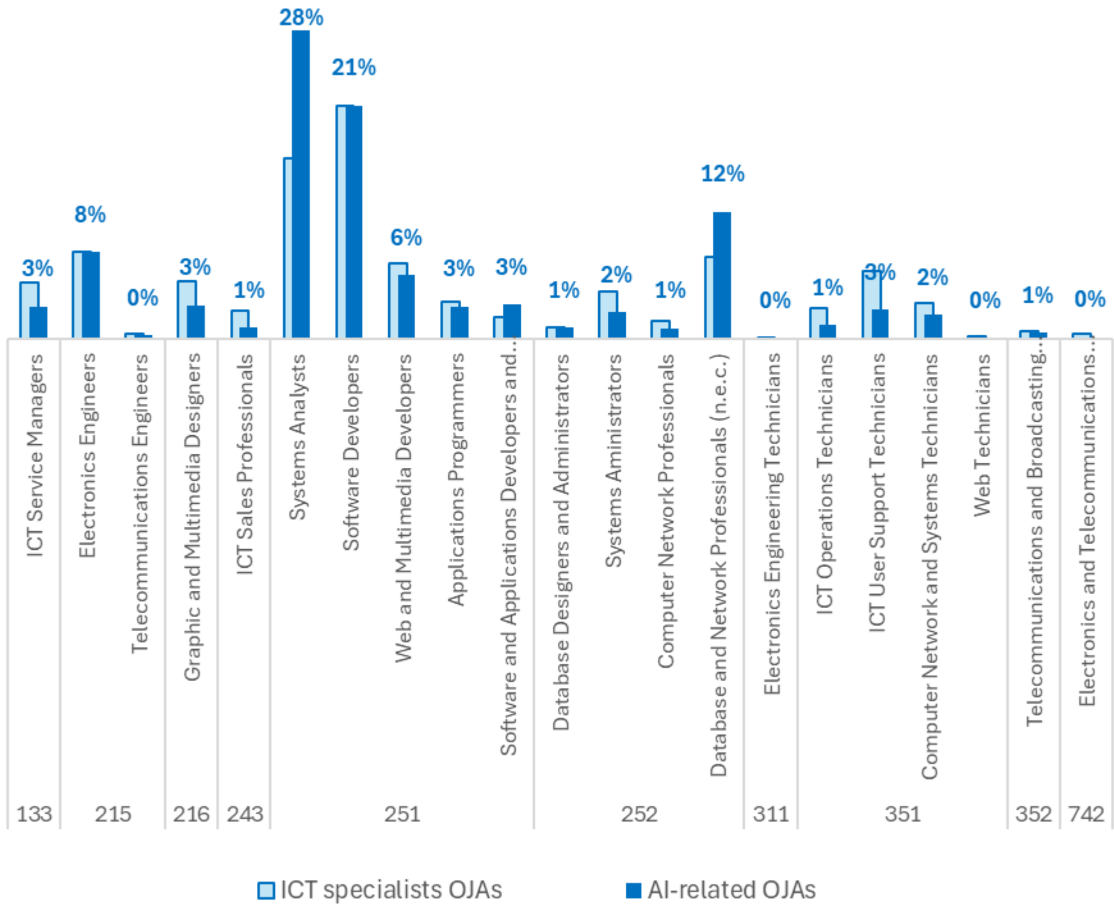
3.2. AI-related demand in online job advertisements

This sub-section describes the main characteristics of the AI-related OJAs considered in the study and presents them in relation to the set of all OJAs that correspond to ICT specialist occupations, in view of showing the specificities of AI-related OJAs. As explained in Section 2, AI-related OJAs are defined by those whose descriptions include any of the AI keywords used in the study (see Annex 1). The analysis is performed on a representative sample of all OJAs from the reference period 2020-2023, across EU-27, with keywords in the English language.

3.2.1. Analysis by ICT specialists occupations

Figure 5 shows the distribution of AI-related OJAs (dark blue bars) and all OJAs (light blue bars) across ICT specialists occupations. The distribution of AI-related OJAs is generally similar to that of all OJAs for ICT specialists, with few notable differences. Results show **that 62% of AI-related OJAs are in occupation group *251 - Software and Applications Developers and Analysts***, **which accounts for half of all ICT specialist OJAs**. In this group, AI OJAs are particularly highly concentrated in *Systems Analysts* (28%) and *Software Developers* (21%). A further 12% of AI-related OJAs correspond to *Database and Network Professionals (not elsewhere classified (n.e.c.))* in occupation group *252 - Database and Network Professionals*. The assignment of 'not elsewhere classified' is made to occupations which do not fit under existing occupational categories of ISCO, and hence includes some emerging occupations (Cedefop, 2023). A comparison of the dark and light blue bars indicates that there is a higher-than-average concentration of AI-related OJAs in several occupations.

Figure 5. Distributions of AI-related OJAs (dark blue bars) and of all OJAs targeting ICT specialists (light blue bars) by occupation (%), EU 2020-23



Source: JRC's elaboration based on Cedefop's extraction from WIH-OJA data.

Note: The underlying data is a representative sample of all OJAs in EU27 in the reference period 2020-2023, with keywords in the English language. See Section 2.2 for the definition of AI-related OJAs.

These results suggest a specialisation pattern of AI-related OJAs, with some ICT specialist occupations playing a more prominent role as they are over-represented within AI-related OJAs in comparison with all OJAs. So, we explore the extent of AI specialisation in each ICT specialist occupation through an '*AI specialisation index*', defined as *the ratio of the share of AI OJAs in an occupation to the share of all OJAs targeting ICT specialists in that occupation*. Index values exceeding 1 indicate that AI-related OJAs are relatively highly concentrated in an occupation than the overall OJAs, while values lower than 1 indicate that AI-related OJAs are relatively less concentrated in an occupation group.

Figure 6. AI specialisation Index of ICT specialist occupations, EU 2020-23

Source: JRC's elaboration based on Cedefop's extraction from WIH-OJA data.

Note: The AI specialisation index is the ratio of the share of AI-related OJAs for each occupation to the share of all OJAs for that occupation. The underlying data is a representative sample of all OJAs in EU27 in the reference period 2020-2023, with keywords in the English language. Index values exceeding 1 indicate that AI-related OJAs are relatively highly concentrated in an occupation than the overall OJAs, while values lower than 1 indicate that AI-related OJAs are relatively less concentrated in an occupation group. See Section 2.2 for the definition of *AI-related OJAs*.

Figure 6 indicates that three occupations show a **higher AI specialisation – *Systems Analysts*** (1.71 times more frequent among AI-related OJAs than in overall OJAs); ***Database and Network Professionals*** *(n.e.c.)* (1.56 times more frequent); and ***Software and Applications Developers and Analysts*** *(n.e.c.) (1.55 times more frequent)*. AI specialisation is lowest (less than 0.5) among

Electronics and Telecommunications Installers and Repairers; Web Technicians; *ICT sales professionals*; *ICT User Support Technicians*; and *ICT operations technicians*. AI specialisation is also rather low for *Systems Administrators*; *ICT Service Managers*; *Graphic and Multimedia Designers*; and *Computer Network Professionals*. The results for the AI specialisation index suggest a marked unevenness in the extent to which different ICT specialist occupational vacancies contain AI-related themes.

3.2.2. Analysis of AI job profiles within ICT specialist occupations

As described in Section 2.2, the analysis of OJAs uses the same set of keywords as the analysis of education and training offer (Annex 1, Table A1), plus a set of *AI job profile* keywords that were uniquely applied to OJAs and associated to specific job profile groups (Annex 1, Table A2).

In this analysis, we define *AI job profiles* as *specific roles performed by ICT specialists.* The analysis of the keywords related to AI job profiles mentioned in the description of the job advertisements, although not present in all AI-related OJAs identified, allows us to infer the most demanded AI job profiles, or roles, in the labour market[18]. Such an analysis is necessary because the ISCO standard classification is not directly linked to AI, meaning that AI-related occupations are not explicitly identifiable. While the previous section examined the share of online job advertisements related to AI within each ISCO occupation, this section focuses on the AI profiles that emerge from the text of these AI-related OJAs, and evaluates their distribution and consistency across the ISCO occupations. For instance, the AI profile *AI/ML Engineering* is one of the profiles we search for in the OJAs, and we expect it to be more prevalent in certain ISCO occupations, such as *Software developers*. At the same time, this profile does not align perfectly with the ISCO classification, which is not specifically designed for AI, and therefore the same profile may also appear across other ISCO occupations.

A caveat to consider when interpreting these results (see also Section 2.2) is that a keyword-based match cannot guarantee that the OJA in question belongs to a specific AI profile. For example, two (fictitious) text extracts, 'You will work as a lead Data Analyst' and 'You will work together with our Data Analyst' in a job description would both trigger a match for the Data Analysis profile (role). Therefore, results should be considered as indicative, but not conclusive.

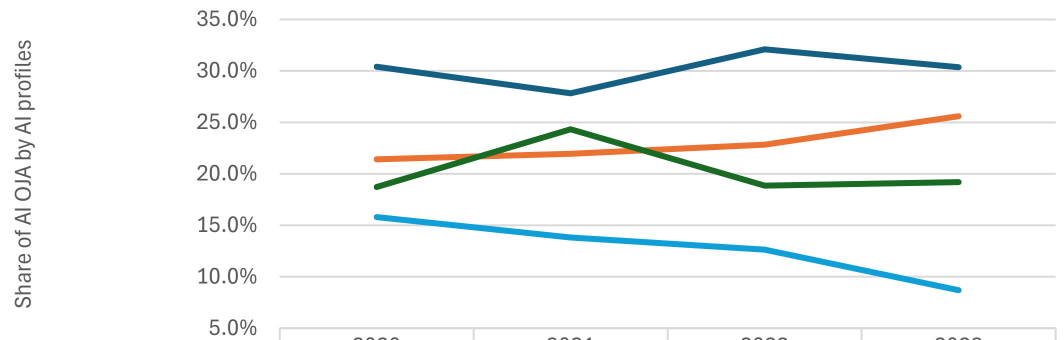
Figure 7 shows the top four AI job profiles, as classified in Table A2. Together, these account for 84% of all job descriptions referencing an AI role. The **most in-demand profile is AI/ML Engineering, representing on average around 30% of postings explicitly mentioning AI profiles**. This share has remained relatively stable between 2020 and 2023, with only minor fluctuations. The **second most prominent category is Data Analysis, accounting for 23%** of demand and showing a consistent upward trend over the period. This growing trend may suggest a shift toward data-driven decision-making, as organisations are more likely to require insights from large datasets. A similar share is observed for Data Engineering roles (around 20%), though unlike Data Analysis, this category has not shown long-term growth—peaking at 24% in 2021 but stabilising afterwards. The stable trend of this job profile may indicate a more mature, consolidated role. In contrast, demand for AI/ML Development declined from nearly 14% in 2020 to 12% in 2023. We can only speculate that this decline may be due to efficiency gains, reducing the number

[18] The list of keywords used to detect AI OJAs includes an extensive set of AI job profiles/professions. All OJAs detected contain at least one keyword from the keyword list, however not in all cases a job profile is mentioned.

of developers required, or to an overlap with Software AI/ML Engineers, whose broader skillsets allow them to manage the full process from development to deployment.

Figure 8 reports the share of the other AI job profiles, representing overall the 14% of the AI-related OJAs mentioning AI profiles. We can see that, after the four profiles in **Figure 7**, AI Product & UX Design roles have the largest share, with an average of 4.8%, and show a steady increase from 3.9% to 5.2% between 2020 and 2023. Notably, **AI Governance & Operations, even though it has a low average share, experienced a strong upward trend, rising from 1.5% to 4.2%** in the same period. This could be due to the growing importance of ethical, regulatory, and operational oversight as organisations increasingly integrate AI into their business processes.

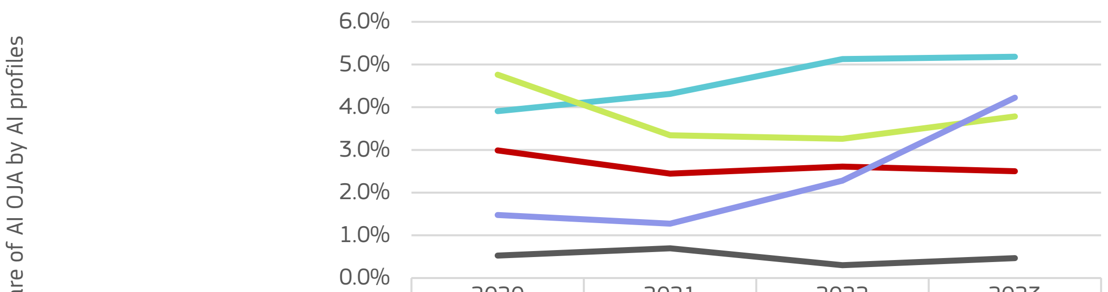
Figure 7. Top 4 AI job profiles in AI OJAs within ICT specialists, EU 2020-23



	2020	2021	2022	2023
AI/ML Engineering	30.4%	27.8%	32.1%	30.4%
Data Analysis	21.4%	22.0%	22.8%	25.6%
Data Engineering	18.7%	24.3%	18.9%	19.2%
AI/ML Development	15.8%	13.8%	12.6%	8.7%

Source: JRC's elaboration based on Cedefop's extraction from WIH-OJA data.

Figure 8. Other AI job profiles in AI OJAs within ICT specialists, EU 2020-23



	2020	2021	2022	2023
AI Product & UX Design	3.9%	4.3%	5.1%	5.2%
AI Research & Applied Science	4.8%	3.3%	3.3%	3.8%
AI Solutions & Architecture	3.0%	2.4%	2.6%	2.5%
AI Governance & Operations	1.5%	1.3%	2.3%	4.2%
Others	0.5%	0.7%	0.3%	0.5%

Source: JRC's elaboration based on Cedefop's extraction from WIH-OJA data.

Note: The figure excludes 2.5% of records containing 'other' job profiles, which fall outside of the top 20. Green indicates an increase in demand; blue indicates that there is no discernible trend or that the demand is stable; and orange indicates a decrease in demand

Figure 9 shows, for a selection of ICT specialist occupations[19], the top AI job profiles in AI-related OJAs (dark blue bars) and all OJAs (light blue bars). These results are important to illustrate how specific AI profiles are distributed across the standard ISCO occupations and to assess their consistency. Indeed, the predominance of certain AI job profiles in each occupation highlights differential patterns of job profiles for the time span analysed (2020-23).

Systems Analysts, which represents the most common ICT occupation for AI-related OJAs (see **Figure 5**), has a strong concentration of Data Analysis profiles (39%) - a share considerably higher than the average across all ICT occupations. AI/ML Engineering (26%) and AI/ML Development (17%) also account for a large proportion, highlighting the presence of highly technical profiles related to machine learning and artificial intelligence.

For the Database and Network Professionals occupation, the predominant AI profile is Data Engineering, which represents 57% of all AI-related profiles. The emphasis on Data Engineering may suggest the demand for expertise in managing large-scale data flows, ensuring data quality, and integrating heterogeneous data sources.

For Software Developers, Electronics Engineers[20], and Application Programmers occupations, the defining feature is the high prevalence of AI/ML Engineering, accounting for 53%, 60%, and 63% respectively. Therefore, the strong concentration of AI/ML Engineering in **Figure 7** appears to be driven primarily by these three occupations.

For the Graphic and Multimedia Design occupation, most AI profiles (70%) relate to AI Product & UX Design, a share substantially higher than its modest representation in the overall distribution (below 5% among all OJAs targeting ICT specialists). This suggests that in creative ICT professions, demand for AI profiles is driven primarily by the development of user-centred applications and interfaces rather than engineering or data-related tasks.

For the ICT Operations Technicians occupation, AI profiles vary. The largest shares are for Data Analysis (31%) and Data Engineering (25%), but smaller, non-negligible proportions are visible for AI Governance & Operations (18%)—a category which is nearly absent in most other ICT roles. This indicates that AI adoption in operations support is linked to oversight, monitoring, and governance functions.

Finally, among Database Designers and Administrators occupations, AI profiles are almost equally split between AI/ML Development (34%) and Data Engineering (32%), whereas across ICT specialist occupations in general, these categories account for smaller shares. This suggests that AI adoption in database professions is particularly focused on development and infrastructure optimisation.

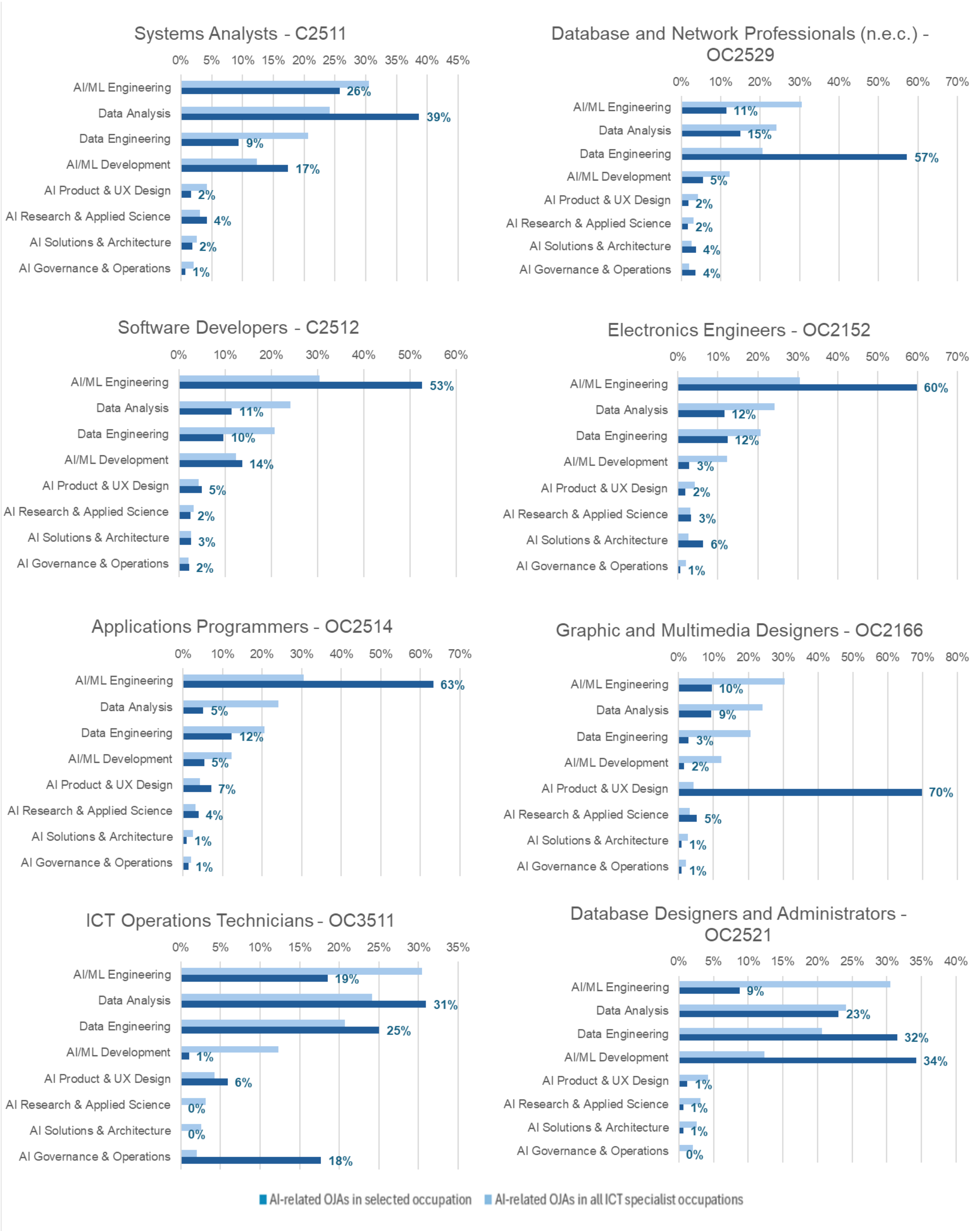
Overall, the **analysis shows that while AI/ML Engineering roles have a predominant position across many ICT specialist occupations**, there are important deviations by occupation. **Design roles emphasise AI in product and user experience**, **systems analysis is more data-centric, database roles are split between AI engineering and AI development profiles, and**

[19] The figure includes the eight most targeted ICT specialists occupations for which the number of OJAs including an AI-profile keyword is above 300.

[20] We should acknowledge that the high presence of AI/ML Engineering within this occupation may partly reflect the way our keyword search categorises AI-related activities, which can result in different types of AI engineering being associated with Electronic Engineers.

operations staff are more engaged with AI governance profiles. This occupation-specific variation suggests that AI integration in ICT professions is both broad-based and highly differentiated, reflecting the diverse ways AI technologies are being applied across the digital economy.

Figure 9. Top AI job profiles in AI-related OJAs by ICT specialist occupation, EU 2020-23



Source: JRC's elaboration based on Cedefop's extraction from WIH-OJA data.

Note: The figures show the distribution of AI profiles across the eight ICT occupations selected, which are ordered in descending order according to the share of AI OJA relative to the total job ads in each occupation. The dark blue bars represent the distribution of AI profiles within each specific occupation, while the light blue bars are reported for comparison and remain the same across all figures, as they show the distribution of AI profiles in all ICT occupations.

3.3. Comparing AI-related education and training offer with OJAs demand

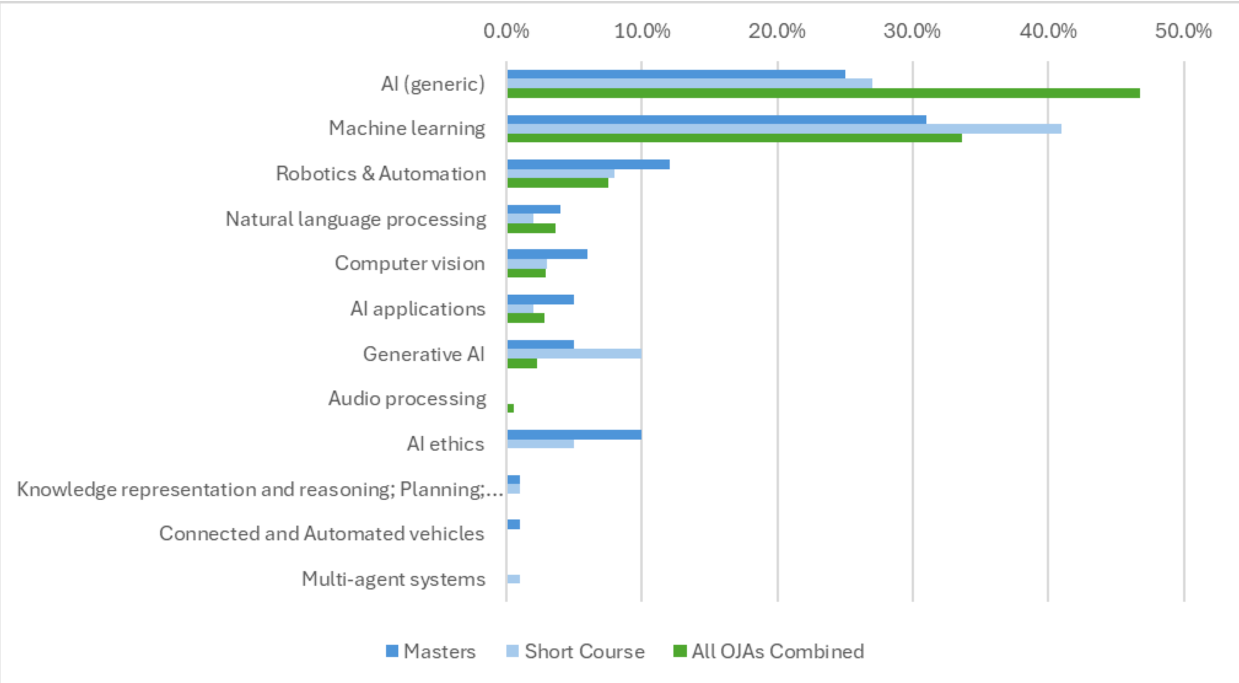
This section offers a broad comparison between AI-related education and training offer and labour market demand (in OJAs), using the AI subdomains as a basis. As noted in Section 2, the keywords have been grouped in accordance with a taxonomy of AI (Annex 1), which provides useful information for the analysis of both education and training programmes and OJAs.

Figure 10 compares the percentages of masters' programmes and short courses falling into each of the AI subdomains (dark and light blue bars, respectively) with the percentages of AI OJAs for ICT specialists which can be attributed to the same AI subdomains (green bars).

It should be noted that the reference period for the two data sources is not the same: data for education and training offer covers the academic year 2024-2025 while the OJAs cover the period 2020-2023. Therefore, **Figure 10** is a comparison of recent AI-related demand among ICT specialist occupations with current and forthcoming offer. The caveats mentioned earlier in the report of course also apply here, meaning that these comparisons should be interpreted in a broad way and considered alongside other sources of information and evidence.

In general**, offer and demand align quite closely, although there are some notable differences**. For example, the offer-demand differences in relation to generic AI and generative AI could well relate to the increasing proliferation of generative AI since 2023. Furthermore, the appearance of AI ethics in course offerings, which is absent from the OJAs 2020-2023, could reflect an increasing awareness and concern with this aspect of AI systems in educational curricula. Also, AI ethics could be a role that may be less commonly assigned to ICT specialist occupations.

Figure 10 10. Distribution of AI-related education and training offer (2024-2025) and AI-related ICT specialist OJAs (2020-23) by AI subdomain (%), EU



Source: JRC's elaboration based on Studyportals and WIH-OJA data.

4. Conclusions

This report provides an overview of the education and training offer on AI (in particular, of master's degrees and short courses), as well as AI-related labour demand (from an analysis of OJAs targeting ICT specialist occupations). The analysis is concerned with all manifestations of AI, including more specific techniques and technologies such as generative AI or machine learning. For both analyses, a keyword-based identification methodology was employed: this permits a thematic examination of AI-related offer and demand. It should be noted that the analysis is restricted to the English language in both sources (Studyportals for offer, and WIH-OJA for demand), and the time period for the demand data (2020-2023) precedes the time period (academic year 2024-2025) for the offer data. Additionally, it must be considered that OJAs, albeit coming in large volumes and in real-time, are not necessarily representative of the whole ICT specialists labour market. Nonetheless, some insights and preliminary conclusions can be drawn.

The analysis of master's offer shows that content is primarily concentrated on Machine Learning (36%), generic AI (26%), Robotics and Automation (13%) and AI ethics (10%) while Generative AI constitutes less than 1% of programmes' content. AI-related specialised (as opposed to broad) programmes are relatively more common in short courses than in master's programmes, but in both cases, they constitute more than half of all programmes in each level. Short courses also include a higher share of their content on Machine Learning (42%) and generic AI (27%) topics, Robotics and Automation (8%) and AI ethics (5%).

As might be expected, AI-related programmes tend to be taught in the study field of Information and Communication Technologies. Indeed, a high proportion of all ICT programmes cover AI (44%), reaching 11% in Engineering, manufacturing and construction; and 6% among Natural sciences, mathematics and statistics. The much lower penetration rates in fields like Social sciences, journalism and Information; Education; Health and welfare; Agriculture could indicate a time-lag between the growing societal and economic relevance of AI in these fields and their current low integration into both master's and short courses.

Looking at the demand side, about 62% of AI-related OJAs are in the *Software and Applications Developers and Analysts* occupation group (listed by ISCO as category 251). A further 16% of AI-related OJAs are within the occupation group *Database and Network Professionals* (ISCO category 252), and the remaining 22% of AI-related OJAs are dispersed across the remainder of occupations studied.

Based on an occupation's 'AI specialisation index', defined as the ratio of the share of AI OJAs in an occupation to the share of all OJAs targeting ICT specialists in that occupation, AI specialisation is highest among OJAs for *Systems Analysts*; *Database and Network Professionals (n.e.c.)*; and *Software and Applications Developers and Analysts (n.e.c.)* (recall that n.e.c. refers to individual occupations that are not classifiable under other occupations, and include emerging occupations). In contrast, AI specialisation is very low among ICT specialist occupations relating to technician, sales and installation and repair roles, as these occupations are much less frequent among AI-related OJAs than across all ICT specialists' OJAs.

An analysis of information on the profiles or roles related to AI in the OJAs shows that AI/ML Engineering holds a predominant position across many ISCO ICT occupations. However, notable differences emerge by occupation. For instance, graphic and design occupations emphasise AI in product user experience, whereas AI data profiles are particularly prominent in database occupations.

On the basis of the two data sources considered, offer and demand generally align quite closely in what concerns the AI subdomains addressed, although, given the different time periods of the data, results should be interpreted as a comparison of recent AI-related demand with current and forthcoming offer. For example, the offer-demand differences in relation to generic AI and generative AI (where generative AI appears in training offer but not in OJAs) could relate to the increasing proliferation of generative AI since 2023. Similarly, the appearance of AI ethics in course offerings, which is absent from the OJAs, may reflect an increasing awareness of and concern with this theme in education, not necessarily reflected in the labour market, at least among ICT specialists.

In conclusion, the overall broad alignment between offer and demand in relation to AI is to be welcomed. However, the low penetration rates of AI topics in many 'non-ICT' fields of education – both master's degrees and short courses – should be monitored to establish if the potential time-lag in integrating AI topics into various fields may give rise to supply difficulties in the near future. Also, while the inclusion of AI ethics in master's and short courses is to be welcomed given the general importance of this topic, further research could examine the extent to which various courses represent a balance between conceptual, technical and ethical/societal aspects; and it would seem important to monitor the extent to which AI ethics is appearing in OJAs, maybe under a different set of occupations.

Finally, the analysis of AI in OJAs could be extended to a wider range of occupations that may have been already penetrated by AI and generative AI in order to examine patterns of AI-related demand. This could include occupations related to architecture, audiovisual sector (film, television, video games), law, design, visual arts, music and literature creation, cultural heritage (museums, libraries), among others. In this way, future analyses could monitor the integration of AI technologies into strategic sectors beyond ICT, which is particularly important for enhancing quality and efficiency, especially within public sector services. The methodology developed for this study is a step toward in monitoring AI-related offer and demand. It could be repeated over time to complement other efforts to understand AI competence needs and requirements of the workforce.

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List of acronyms

Acronym	Full term
AI	Artificial Intelligence
EU	European Union
ICT	Information and Communication Technologies
ISCO	International Standard Classification of Occupations
JRC	Joint Research Centre (of the European Commission)
N.E.C.	(An individual occupation that is) Not Elsewhere Classified (under the ISCO framework)
OJA(s)	Online Job Advertisement(s)
WIH	Web Intelligence Hub (of the European Commission)

List of figures

List of tables

Annexes

Annex 1. Keyword list

Table A 1. Keywords used in both Education and training offer analysis and OJA analysis

Category	Keyword	Category	Keyword
AI (generic)	artificial intelligence		advanced generative model
	ai application		ai creativity
	business intelligence		artbreeder
	computational creativity		artificial image creation
	computational neuroscience		automated text creation
AI applications	intelligence software		biggan
	intelligent control		claude ai
	intelligent software development		claude.ai
	intelligent system		content creation
	accountability		content generation
	ethics		controlnet
AI ethics	explainability		creative ai
	fairness		creative algorithm
	safety		creative artificial intelligence
	security		creative content generation
	audio processing		creative generation
	speech recognition		creative imaging
	voice recognition		creative text generation
	computer vision	Generative AI	creative writing
Audio processing	face recognition		cyclegan
	facial recognition		dall e
	image classification		dall·e
	image processing		dall-e
	image recognition		deep dream
Connected and Automated vehicles	autonomous system		deep generative models
	expert system		deepai
	inductive programming		deepfloyd
Knowledge representation and reasoning; Planning; Searching; Optimisation	knowledge reasoning		dreambooth
	knowledge representation		elicit
	knowledge representation and reasoning		forefront ai
	uncertainty		gaugan
	adaptive learning		gemini ai
	anomaly detection		generative adversarial network
	deep learning		generative algorithm
Machine learning	deep neural network		generative art
	hyperparameter fine tuning		generative content
	hyperparameter fine-tuning		generative model
	hyperparameter tuning		generative neural network

Category	Keyword
	machine learning
	model validation
	neural network
	parameter efficient fine tuning
	parameter efficient fine-tuning
	parameter tuning
	parameter-efficient fine tuning
	pattern recognition
	recommendation engine
	recommendation platform
	recommendation system
	recommender system
	reinforcement learning
	supervised learning
	support vector machine
	transfer learning
	unsupervised learning
Multi-agent systems	intelligent agent
	multiagent system
Natural language processing	automatic translation
	chat bot
	chatbot
	chat-bot
	chatbot ui
	computational linguistics
	language modeling
	lexical token
	machine translation
	natural language processing
	natural language understanding
	semantic modeling
	semantic search
	text miner
	text mining
	tokenization
	word embeddings
	word tokenization
Robotics & Automation	control theory
	human-ai interaction
	robotics
	gigagan
	gpt
	hugging face
	image creation
	image generation
	innovative creation
	lamda
	langchain
	language generation
	language generation model
	large language model
	lexical tokenization
	llama
	llm
	media generation
	midjourney
	mistral ai
	mixtral
	mixtral of expert
	multimodal generation
	natural language generation
	openai
	opentts
	perplexity
	prompt engineer
	prompt engineering
	rag
	retrieval-augmented generation
	runway ml
	stylegan
	suno ai
	text generation
	text-to-image generation
	vector representations
	visual generation
	visual synthesis
	voice.ai
	zephyr

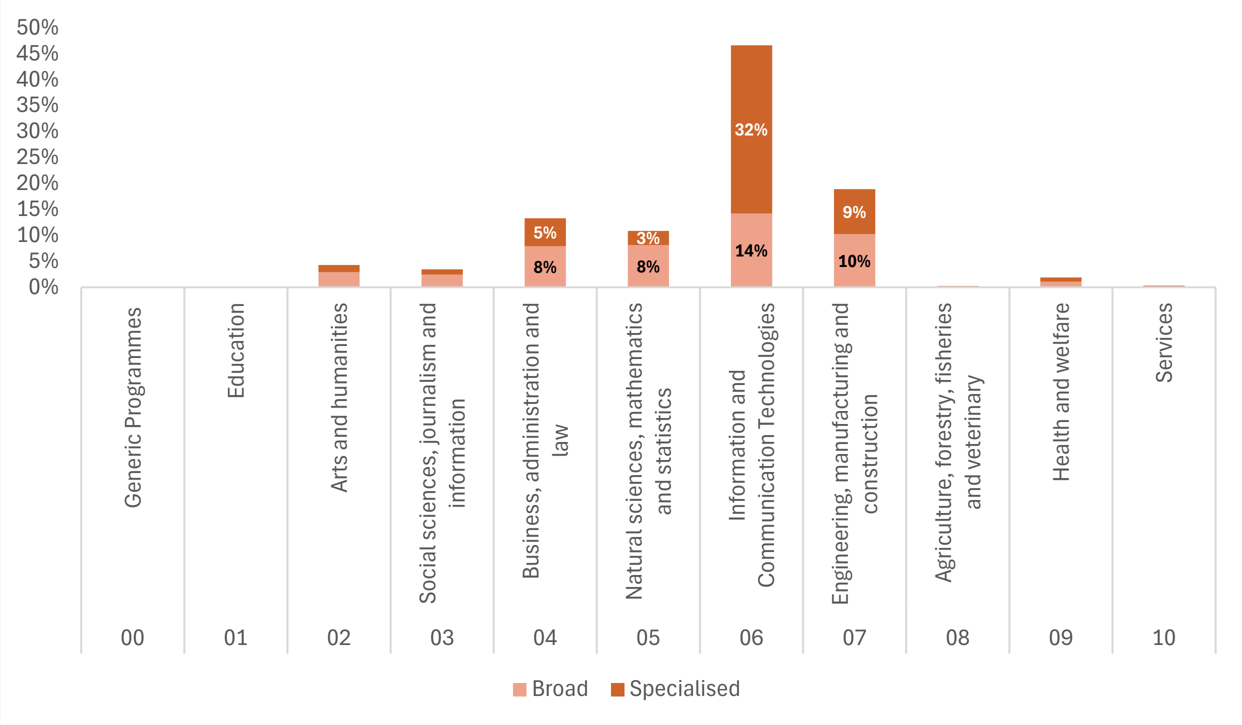
Table A 2. List of AI profiles keywords used in OJA analysis only

Profile Keywords	Classification Group
AI Operations Manager	AI Governance & Operations
Compliance Officer	AI Governance & Operations
Risk Manager	AI Governance & Operations
Security Specialist	AI Governance & Operations
AI Product Manager	AI Product & UX Design
Conversational Designer	AI Product & UX Design
Product Manager	AI Product & UX Design
UX Designer	AI Product & UX Design
UX Developer	AI Product & UX Design
AI Research Scientist	AI Research & Applied Science
Applied Scientist	AI Research & Applied Science
Research Engineer	AI Research & Applied Science
Research Scientist	AI Research & Applied Science
Researcher	AI Research & Applied Science
AI Platform Engineer	AI Solutions & Architecture
AI Solutions Architect	AI Solutions & Architecture
Solutions Architect	AI Solutions & Architecture
Solutions Engineer	AI Solutions & Architecture
AI Trainer	AI Training & Annotation
AI Developer	AI/ML Development
Algorithm Developer	AI/ML Development
Business Intelligence Developer	AI/ML Development
Chatbot Developer	AI/ML Development
AI Engineer	AI/ML Engineering
Computer Vision Engineer	AI/ML Engineering
Deep Learning Engineer	AI/ML Engineering
Deep Learning Specialist	AI/ML Engineering
Generative AI Engineer	AI/ML Engineering
Machine Learning Engineer	AI/ML Engineering
Modeling Engineer	AI/ML Engineering
Natural Language Processing Engineer	AI/ML Engineering
NLP Engineer	AI/ML Engineering
Prompt Engineer	AI/ML Engineering
Software Engineer	AI/ML Engineering
AI Data Analyst	Data Analysis
Big Data Analyst	Data Analysis
Data Analyst	Data Analysis
Data Mining and Analysis	Data Analysis
Data Scientist	Data Analysis
Big Data Architect	Data Engineering
Big Data Engineer	Data Engineering
Data Engineer	Data Engineering
AI Consultant	Executive Leadership
Chief AI Officer	Executive Leadership
Quality Assurance Tester	Quality & Testing
Hardware Specialist	Robotics & Hardware
Robotics Engineer	Robotics & Hardware

Source: JRC's elaboration.

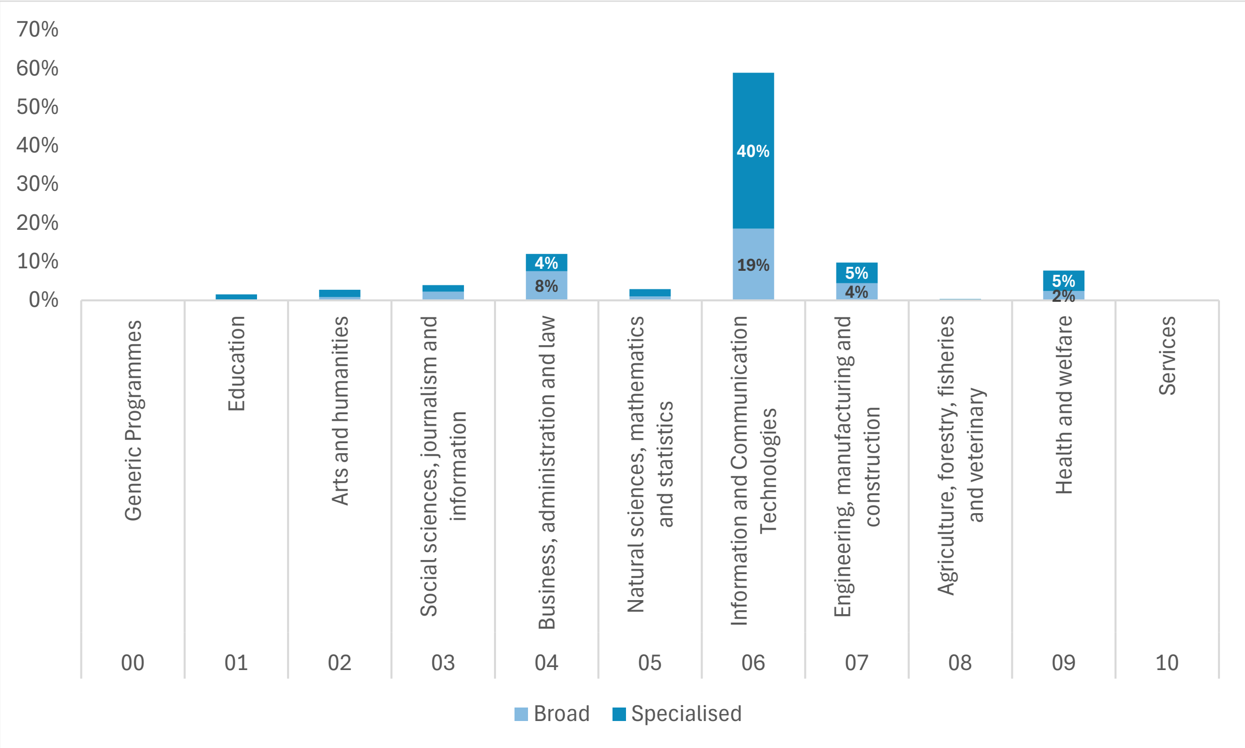
Annex 2. Detailed results

Figure A 1. Distribution of AI Master's programmes by field of study (%) and Scope, EU 2024-2025



Source: JRC's elaboration based on Studyportals data.

Figure A 2. Distribution of AI Short Courses' programmes by field of study (%) and Scope, EU 2024-2025



Source: JRC's elaboration based on Studyportals data.

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